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## Exercise 12

Due date: **Tuesday, May 26<sup>th</sup> 2026, 14:00**

### Submission Notes:

- Please submit your answers using the corresponding function in ADAM.
- Write down your solutions in a way that your reasoning and solution path is understandable.
- Please submit your solutions to the coding tasks as executable code and make sure that your implementations are working. If you encounter problems and cannot solve them, add additional notes and comments to your solution about where you think the error is or what you have tried and tested so far.
- Plagiarism results in an automatic fail for exercise sheets, i.e. every student needs to hand in their own solution.

### 12.1 Probabilistic PCA

Recall the mixture models, which use a single latent variable to generate the observations, i.e. each observation can come from one of  $K$  clusters. In other words, the model uses  $K$  hidden binary variables representing the cluster identity. However, the representational power of the model is limited due to mutually exclusive latent variables. Instead, one can use a vector of real-valued latent variables  $\mathbf{z}_i \in \mathbb{R}^K$ . For continuous observations, a Gaussian for the likelihood can be assumed, such that

$$p(\mathbf{x}_i | \mathbf{z}_i, \boldsymbol{\theta}) = \mathcal{N}(\mathbf{W}\mathbf{z}_i + \boldsymbol{\mu}, \boldsymbol{\Psi}),$$

where  $\mathbf{W} \in \mathbb{R}^{D \times K}$  is orthonormal,  $\boldsymbol{\Psi} \in \mathbb{R}^{D \times D}$  is a diagonal covariance matrix  $\boldsymbol{\Psi} = \sigma^2 \mathbf{I}$  and  $\boldsymbol{\mu}$  is the mean of inputs  $\mathbf{X}$ . If  $\sigma \rightarrow 0$ , then the model reduces to the classical principal components analysis. In order to find orthonormal set of  $K$  linear basis vectors  $\mathbf{W}$  and the corresponding scores  $\mathbf{Z}$ , the following objective function

$$J(\mathbf{W}, \mathbf{Z}) = \|\mathbf{X} - \mathbf{W}\mathbf{Z}^T\|_F^2$$

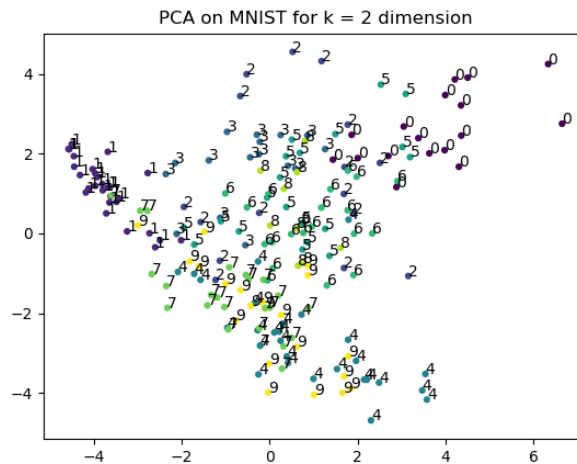


Figure 1: A plot of a two-dimensional representation of the MNIST images. Note that similar classes, such as ones and zeros are rather clustered.

has to be minimized, where  $\|\mathbf{A}\|_F = \sqrt{\text{tr}(\mathbf{A}^T \mathbf{A})}$  is the Frobenius norm. The optimal solution is obtained if  $\mathbf{W}$  contains  $K$  eigenvectors with largest eigenvalues of the empirical covariance matrix  $\hat{\Sigma} = \frac{1}{N} \sum_{i=1}^N \mathbf{x}_i \mathbf{x}_i^T$ . Hence, the optimal low-dimensional encoding

$$\mathbf{Z} = \mathbf{X}\mathbf{W}$$

is an orthogonal projection of the data onto the column space spanned by the eigenvectors.

### Exercise

One can obtain the solution by using singular value decomposition (SVD). To do so, decompose the data  $\mathbf{X} = \mathbf{U}\mathbf{S}\mathbf{V}^T$ , where  $\mathbf{U} \in \mathbb{R}^{N \times N}$  with orthonormal columns  $\mathbf{U}^T \mathbf{U} = \mathbf{I}_N$ ,  $\mathbf{V} \in \mathbb{R}^{D \times D}$  with orthonormal rows and columns  $\mathbf{V}^T \mathbf{V} = \mathbf{V}\mathbf{V}^T = \mathbf{I}_D$  and  $\mathbf{S} \in \mathbb{R}^{N \times D}$  containing singular values on the diagonal. Note that  $\mathbf{X} \approx \mathbf{U}_{:, :K} \mathbf{S}_{:K, :K} \mathbf{V}_{:, :K}^T$ . The linear projection to  $K$ -dimensional space is obtained by setting  $\mathbf{W} = \mathbf{V}_{:, :K}^T$ .

Write a Python program to identify  $K$  principal components on the MNIST dataset.

- Load the MNIST dataset and compute SVD on the image data with the `numpy` package.
- Choose  $K = 2$  principal components with largest eigenvalues and project  $\mathbf{X}$  to a  $K$ -dimensional representation  $\mathbf{X}_{enc}$ .
- Decode  $\mathbf{X}_{enc}$  back to the original space, i.e compute  $\mathbf{X}_{dec} \approx \mathbf{X}$ .
- Plot the two-dimensional scatter of the latent representation as shown in Figure 1 and the corresponding reconstructs  $\mathbf{X}_{dec}$  as shown in Figure 2.
- Try out different settings for  $K$ . What do you notice?

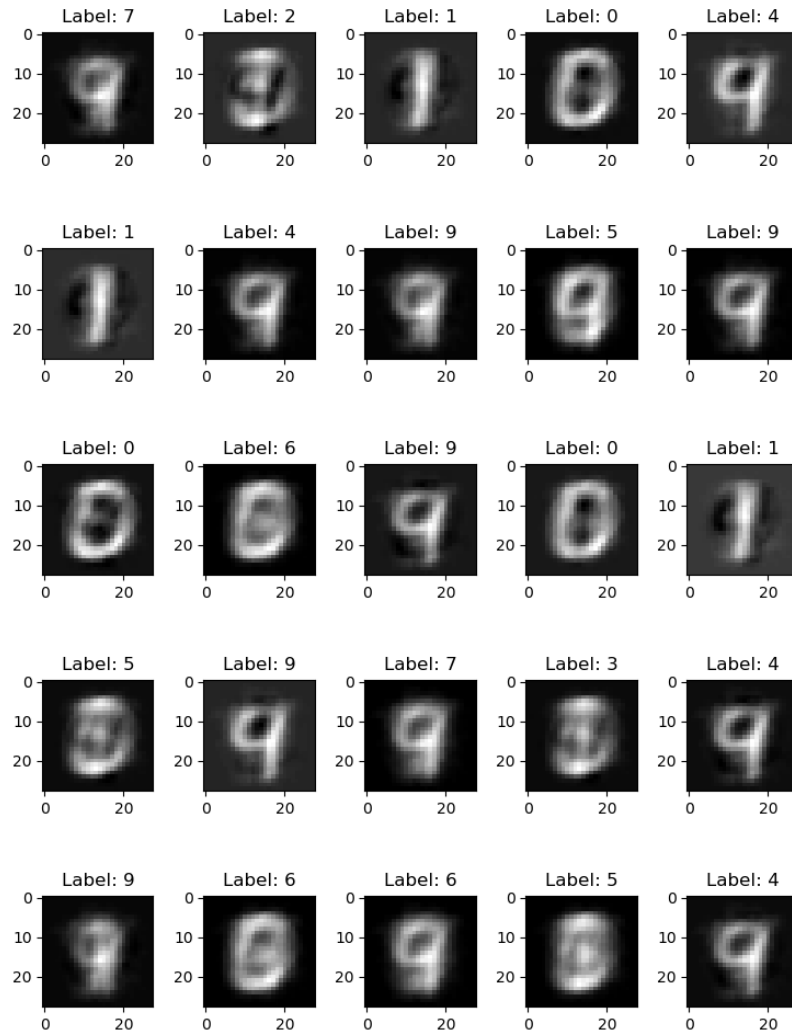


Figure 2: A plot of the decoded MNIST images from a very low two-dimensional representation. The reconstructions are rather poor, however for the *ones* it seems to work already reasonably well.

## 12.2 [BONUS] Variational AutoEncoder (VAE)

For the last exercise, you will face a little challenge. This task is a BONUS task, which means you are not strictly required to solve it, but highly encouraged. And you would miss out on a very fascinating approach.

Another way to achieve dimension reduction is the usage of Variational Autoencoders (VAEs). These neural networks consist of two parts: an encoder, which transforms some input into a latent representation, often modeled as a (multivariate) normal distribution with mean and standard deviation; and a decoder, which aims to reconstruct the original signal from the latent representation.

Try to implement and train a VAE on the MNIST dataset. This time, you are free to use existing implementations, study available tutorials, or come up with your very own solution. There is no specific template for this task, but you can use whatever you have seen in exercise 12.1 as well as for the neural network sessions. Use the encoding part to plot and compare the latent representation provided by the VAE to the result of the PCA. You can also make use of the decoder to explore the latent space, walk (randomly or structured) through it and reconstruct images from the latent embeddings. What do you notice? How would you compare PCA and VAEs?